IC Engines Tutorial

Questions

1. A four stroke gas engine has a bore of 20 cm and stroke of 30cm and runs at 300 rpm firing every cycle. If air fuel ratio is 4:1 by volume and volumetric efficiency on NTP basis is 80%, Determine the volume of gas used per minute. If the calorific value of the gas is 8 MJ/m^3 at NTP and brake thermal efficiency is 25% determine the brake power of the engine. (Ans. 0.2262 m^3 at NTP/min. bp=7.54 kW)

2. The following observations have been made from the test of a four cylinder two stroke gasoline engine. diameter 10cm, stroke 15cm, speed 1600 rpm. Area of the positive loop of the indicator diagram 5.75 sq.cm., are of the negative loop of the indicator diagram 0.25 sq.cm. Length of the indicator diagram 55mm. Spring constant 3.5 bar/cm. Find the indicated power of the engine. (Ans. 43.98 kW)

3. A single cylinder engine running at 1800 rpm develops a torque of 8 Nm. The indicated power of the engine is 1.8 kW. Find the loss due to friction as the percentage of brake power. (Ans 16.22 %)

4. A gasoline engine working on Otto cycle consumes 8 liters of gasoline per hour and develops 25 kW. The specific gravity of gasoline is 0.75 and its calorific value is 44000 kJ/kg. Determine the indicated thermal efficiency of the engine (ans. 34.1 %)

5. A four cylinder engine running at 1200 rpm delivers 20kW. The average torque when one cylinder was cut is 110 Nm. Find the indicated thermal efficiency if the calorific value of the fuel is 43 MJ/kg and the engine uses 360 gm gasoline per kWhr. (Ans. 28.74 %)

6. The bore and stroke of a water cooled single cylinder four stroke diesel engine are 80mm and 110 mm respectively and the torque is 23.5 Nm. Calculate the brake mean effective pressure of the engine.(Ans. 5.34 bar)

7. Find the brake specific fuel consumption in kg/kWh of a diesel engine whose fuel consumption is 5 gm /s when the power output is 80 kW. If the mechanical efficiency is 75% calculate the indicated specific fuel consumption. (Ans. bsfc 0.225 kg/kWh, isfc = 0.169 kg/kWh)

8. Find the air fuel ratio of a four stroke single cylinder air cooled engine with fuel consumption time for 10 cc as 20.4 s and air consumption time for 0.1 m^3 as 16.3 s. The load is 17kg at a speed of 3000 rpm. Find also brake specific fuel consumption in g/kWh and brake thermal efficiency. Assume the density of air as 1.175 kg/m^3 and specific gravity of fuel 0.7. The calorific value of

fuel 43 MJ/kg and dynamometer constant is 5000. (Ans air fuel ratio 21, BP = 4.2 kW, bsfc 294 g/kWh,η_{bth}=28.48%)

9. A six cylinder gasoline engine operates on the four stroke cycle. The bore of each cylinder is 80 mm and the stroke 100 mm. The clearance volume per cylinder is 70 cc. At a speed of 4000 rpm, the fuel consumption is 20 kg/h and the torque developed is 150 Nm. Calculate brake power, brake mean effective pressure, brake thermal efficiency. The calorific value of fuel 43 MJ/kg. Relative efficiency on a brake power basis assuming the engine works on the constant volume cycle. Take $\gamma = 1.4$ for air. (Ans. bp = 62.8 kW, P_{bm}=6.25 bar, η_{bth}=26.3%, r=8.18, $\eta_{otto} = 56.8\%$. $\eta_{rel} = 46.3\%$.)

10. An eight cylinder four stroke engine of 9cm bore and 8 cm stroke with a compression ratio of 7 is tested at 4500 rpm on a dynamometer which has 54cm arm. During a 10 minutes test the dynamometer scale reading was 42 kg and the engine consumed 4.4 kg of gasoline with CV 44 MJ/kg. Atmospheric air at 27 °Cand 1 bar ism supplied at 6 kg/min. Find bp, brake mean effective pressure, brake thermal efficiency, bsfc, brake specific air consumption, volumetric efficiency, air fuel ratio. (Ans. bp=104.8 kW, bmep = 6.87 bar, bsfc =0.252 kg/kWh, bsac =3.435 kg/kWh, η_{bth}=32.5%, η_v=56.44%, Air fuel ratio =13.64)

11. The air flow to a four cylinder four stroke oil engine is measured using an orifce meter with dia 5cm, and c_d=0.6. The engine bore is 10cm, stroke 12cm, speed =1200 rpm, brake torque 120 Nm, fuel consumption 5kg/h, CV of fuel 42 MJ/kg, pressure drop across orifce is 4.6 cm of water , ambient is at 17 °Cand 1 bar. Calculate brake thermal efficiency, brake mean effective pressure, volumetric efficiency. (ans. bp= 15.08 kW, η_{bth}=25.85%,P_{bm}=4 bar,η_v=85.7%)

12. A gasoline engine working on four stroke developes brake power of 20.9 kW. A morse test was conducted on this engine and brake power obtained when each cylinder was made inoperative by short circuiting the spark plug are 14.9, 14.3, 14.8 and 14.5 respectively. The test was conducted at constant speed. Find the indicated power, mechanical efficiency and bmep when all the cylinders are firing. Bore of the engine 75mm, stroke 90 mm, and speed 3000 rpm. (ans IP 25.1 kW, $\eta_{mech} = 83.3\%$., P_{bm}=5.25 bar)

13. The observations of a retardation test on a single cylinder diesel engine are as follows, Rated power 8 kW. Calculate frictional power and mechanical efficiency

Sl No	Drop in Speed (rpm)	time for fall-no load (s)	time for fall-50% load (s)
1	$475 \rightarrow 400$	7	2.2
2	$475 \rightarrow 350$	10.6	3.7
3	$475 \rightarrow 325$	12.5	4.8
4	$475 \rightarrow 300$	15	5.4
5	$475 \rightarrow 275$	16.6	6.5
6	$475 \rightarrow 250$	18.9	7.2

14. A four stroke gas engine has a cylinder dia 25cm and stroke 45 cm. The effective dia of the brake drum is 1.6 m. The observations are listed down here

Duration of test	=	40 min
Total revolutions	=	8080
Total explosions	=	3230
Net load on brake	=	90kg
Mean effective pressure	=	5.8 bar
Volume of gas used	=	7.5m^3
Pressure of gas indicated in meter	=	136 mm water gauge
Atmospheric temperature	=	17 ℃
CV of gas	=	19 MJ/m^3 at NTP
Temp rise of jacket water	=	45 ℃
Cooling water supplied	=	180 kg

Find the heat balance for the engine, indicated thermal efficiency, brake thermal efficiency, Assume Atmospheric pressure as 760 mm Hg. (Ans. bp 14.94 kW, Heat supplied = 3396.25 kJ/min, Heat-BP= 896.4 kJ/min, Heat Cooling= 846.5 kJ/min, Unaccounted loss 1653.35 kJ/min, η_{bth}=26.39%, η_{ith}=30.47%)

15. The following observations made for stroke single cylinder gas engine has a cylinder dia 18cm and stroke 24 cm. The effective dia of the brake drum is 1 m. The observations are listed down here

Duration of test	=	30 min
Total revolutions	=	9000
Total explosions	=	4450
Net load on brake	=	40kg
Mean effective pressure indicated	=	5 bar
Volume of gas used at NTP	=	2.4m^3
Volume of air used	=	36m^3
Pressure of air	=	720 mm of Hg
Atmospheric temperature	=	17 ℃
CV of gas	=	19 MJ/m^3 at NTP
Temp rise of jacket water	=	30 ℃
Cooling water supplied	=	80 kg
Density of air at NTP	=	1.29 kg/m^3
temperature of exhaust gas	=	350℃
specific heat of exhaust gas	=	1 kj/kgK
R	=	287 J/kgK

Find the heat balance for the engine, indicated thermal efficiency, mechanical efficiency, (Ans. ip=7.55 kW, bp 6.16 kW, Heat supplied = 1520 kJ/min, Heat-BP= 369.6 kJ/min, Heat Cooling= 334.4 kJ/min, exhaust gas heat 493.5 kJ/min, Unaccounted loss 322.5 kJ/min, η_{ith}=29.8%, η_m=81.6%)